Quantitative Interview Problem Bank

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1 Probability

Problem 1: Coin Toss Game (St. Petersburg Paradox)

Problem
You toss a fair coin. If H, you stop and win \$1. If T, you toss again. If second is H, you stop and win \$2. If second is T, you toss again, etc. Payoff is 2^{n-1}. What is the expected value of this game?

Solution
The game ends on toss k (the first Heads) with probability $(1/2)^k$, and the payoff in that case is 2^{k-1} dollars. The expected value is therefore:

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} P(\text{first H on toss } k) \cdot 2^{k-1} = \sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^k \cdot 2^{k-1} = \sum_{k=1}^{\infty} \frac{1}{2} = \infty.$$

Every term in the sum equals $1/2$, so the series diverges. The expected value is **infinite**.
This is the **St. Petersburg Paradox**. No rational agent would pay an unlimited price for this game, which highlights the distinction between expected monetary value and expected *utility*. Bernoulli's resolution is that utility is logarithmic in wealth, so the expected utility is finite. Alternatively, any real casino has a finite bankroll, which caps the payoff and makes the expected value finite.

Problem 2: Stick Breaking

Problem
Break a stick of unit length at two uniform random points X, Y. Probability they form a triangle?

Hint
Let the two break points be $U_1, U_2 \sim U[0,1]$. The three piece lengths are determined by the order statistics of these points. For three lengths to form a triangle, the **triangle inequality** simplifies to: no single piece can exceed half the total length.
Draw the unit square with axes U_1, U_2 and shade the region where all three triangle inequalities hold. Computing the area of that region gives the probability directly.

Solution
Let the break points be $X, Y \sim U[0,1]$. Define the order statistics $x = \min(X,Y)$ and $y = \max(X,Y)$. The three piece lengths are:

$$L_1 = x, \qquad L_2 = y - x, \qquad L_3 = 1 - y.$$

For three lengths to form a triangle, the sum of any two must exceed the third. Since $L_1 + L_2 + L_3 = 1$, each triangle inequality reduces to "no single piece can be $\geq 1/2$." The three constraints are:

$$x < \frac{1}{2}, \qquad y - x < \frac{1}{2}, \qquad 1 - y < \frac{1}{2}.$$

The last constraint gives $y > 1/2$, and the second gives $y < x + 1/2$.
In the sorted coordinate space $0 \leq x \leq y \leq 1$ (which has total area $1/2$), the valid region is the triangle bounded by $x = 1/2$, $y = 1/2$, and $y = x + 1/2$. This triangle has area $1/8$.
Therefore:

$$P(\text{triangle}) = \frac{1/8}{1/2} = \mathbf{1/4}.$$

Problem 3: Meeting Problem

Problem
Two people arrive at a cafe independently, uniformly over 1 hour. Each waits time w. Prob they meet?

Hint
Model the two arrival times as uniform random variables $X, Y \sim \text{Uniform}(0, T)$. They meet if and only if $|X - Y| \leq w$, where w is the waiting time.

1. Draw the sample space as a $T \times T$ square.
2. Shade the region where $|X - Y| \leq w$ — this is a diagonal band.
3. The probability is the shaded area divided by the total area.

Think of it as "1 minus the two corner triangles" — each triangle has legs of length $(T - w)$.

Solution
Let arrival times be $X, Y \in [0,1]$ (measured in hours). The two people meet if and only if $|X - Y| \leq w$, where w is the waiting time.
The sample space is the unit square. The "non-meeting" region $|X - Y| > w$ consists of two corner triangles, each with legs of length $(1 - w)$. The probability of meeting is therefore:

$$P(\text{meet}) = 1 - 2 \cdot \frac{1}{2}(1-w)^2 = 1 - (1-w)^2 = w(2-w).$$

Plugging in $w = 15$ minutes $= 1/4$ hour:

$$P(\text{meet}) = \frac{1}{4}\left(2 - \frac{1}{4}\right) = \frac{1}{4} \cdot \frac{7}{4} = \frac{7}{16}.$$

Problem 4: Birthday Paradox

Problem
How many people needed for 50% chance of shared birthday?

Hint
It's easier to compute the complement: $P(\text{at least one match}) = 1 - P(\text{all different})$. Write the probability that all n birthdays are distinct as a product of n terms.
For the approximation, use $\ln(1-x) \approx -x$ for small x, which converts the product into a sum you can evaluate in closed form.
The threshold ends up around $n \approx \sqrt{365}$ because the number of *pairs* grows as $\binom{n}{2} \approx n^2/2$ — it's pairwise collisions that matter, not individual birthdays.

Solution
Compute the complement — the probability all n birthdays are distinct:

$$P(\text{unique}) = \prod_{k=0}^{n-1} \left(1 - \frac{k}{365}\right).$$

Taking logs and applying $\ln(1-x) \approx -x$:

$$\ln P(\text{unique}) \approx -\sum_{k=0}^{n-1} \frac{k}{365} = -\frac{n(n-1)}{2 \cdot 365}.$$

So $P(\text{unique}) \approx e^{-n(n-1)/730}$. Setting $P(\text{match}) = 1 - P(\text{unique}) = 0.5$:

$$\frac{n^2}{730} \approx \ln 2 \approx 0.693 \implies n^2 \approx 506 \implies n \approx 22.5.$$

We need **23 people**.

Problem 5: Markov Chain

Problem
A bug moves on a triangle vertices A, B, C. Prob $1/2$ to move to either neighbor. Exp steps to return to A?

Hint
Two approaches work here:

1. **Stationary distribution:** Find π by solving $\pi = \pi P$ with $\sum_i \pi_i = 1$. The expected return time to state i is $1/\pi_i$.
2. **First-step analysis:** Condition on the first transition from A and set up a recurrence for the expected return time. Exploit the symmetry between B and C.

Both methods should give the same answer — try both as a check.

Solution
Method 1: Stationary distribution. By symmetry, $\pi_A = \pi_B = \pi_C = 1/3$. The expected return time to state i is $1/\pi_i$, so $\mathbb{E}[\text{return to } A] = 3$.
Method 2: First-step analysis. By symmetry B and C are equivalent, so let $E_{B\to A}$ be the expected steps to reach A starting from B (or C). From B, we step to one of two neighbors with equal probability:

$$E_{B\to A} = 1 + \frac{1}{2}(0) + \frac{1}{2}E_{C\to A} = 1 + \frac{1}{2}E_{B\to A}$$

where the first term covers reaching A directly and the second covers going to C (which by symmetry has the same expected time as B). Solving: $E_{B\to A} = 2$.
From A, the first step always goes to B or C, so:

$$\mathbb{E}[\text{return to } A] = 1 + E_{B\to A} = 1 + 2 = 3.$$

Problem 6: Dice Sum to 7

Problem
Roll two fair dice. P(sum = 7)?

Solution
The total number of outcomes is $6 \times 6 = 36$. The favorable outcomes that sum to 7 are:

$$(1,6),\ (2,5),\ (3,4),\ (4,3),\ (5,2),\ (6,1)$$

giving 6 favorable outcomes. Therefore:

$$P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}.$$

Intuition: 7 is the most likely sum because it has the most representations. The possible sums range from 2 to 12, and 7 sits at the center of this distribution.

Problem 7: Conditional Dice

Problem
Roll two dice. Given at least one shows 6, P(both are 6)?

Solution
Let A = "at least one die shows 6" and B = "both dice show 6." First compute $P(A)$ via the complement:

$$P(A) = 1 - P(\text{no } 6) = 1 - \left(\frac{5}{6}\right)^2 = \frac{11}{36}.$$

Since $B \subset A$, we have $P(A \cap B) = P(B) = 1/36$. Applying Bayes' rule:

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{1/36}{11/36} = \frac{1}{11}.$$

Common mistake: Answering 1/6 by reasoning "one die is a 6, so the other is 6 with probability 1/6." This ignores the conditioning — the event "at least one 6" is not the same as fixing a specific die to be 6.

Problem 8: Coupon Collector

Problem
Cereal boxes contain 1 of n distinct toys (uniform). Expected boxes to collect all n?

Hint
Break the collection process into **stages**. Define stage i as "going from $i-1$ distinct toys to i distinct toys."

1. When you already have $i-1$ of n types, the probability the next box contains a **new** toy is $(n-i+1)/n$.
2. What is the expected number of trials to get one success when each trial succeeds with probability p? Derive it from first principles.
3. Use **linearity of expectation**: the total expected number of boxes is the sum of the expected lengths of each stage.

Solution
Break the process into stages. Let T_i be the number of boxes needed to go from $i-1$ distinct toys to i distinct toys. When you already have $i-1$ types, each new box is a "new" toy with probability:

$$p_i = \frac{n-i+1}{n}.$$

What is $\mathbb{E}[T_i]$? Each box is an independent trial with success probability p_i. Let $e = \mathbb{E}[T_i]$. On the first box, with probability p_i we succeed (1 box), and with probability $1-p_i$ we fail and are back where we started:

$$e = p_i \cdot 1 + (1-p_i)(1+e) = 1 + (1-p_i)\,e$$

Solving: $p_i\,e = 1$, so $\mathbb{E}[T_i] = 1/p_i = n/(n-i+1)$.
By linearity of expectation, the total expected boxes is:

$$\mathbb{E}[T] = \sum_{i=1}^{n} \mathbb{E}[T_i] = \sum_{i=1}^{n} \frac{n}{n-i+1} = n \sum_{k=1}^{n} \frac{1}{k} = n\,H_n \approx n \ln n.$$

For $n = 50$: $\mathbb{E}[T] \approx 50 \times 4.5 = 225$ boxes.

Problem 9: Gambler's Ruin

Problem
Start with \$k. Each round: +\$1 (prob p) or −\$1 (prob $q = 1-p$). P(reach \$N before \$0)?

Hint
Use first-step analysis: define P_k as the probability of reaching N from state k, condition on the first step to get a recurrence, apply boundary conditions, and solve. The biased case ($p \neq q$) gives exponential differences rather than constant differences — look at ratios instead of differences.

Solution
Let P_k be the probability of reaching N before 0, starting from state k. Conditioning on the first step:

$$P_k = p \cdot P_{k+1} + q \cdot P_{k-1}$$

with boundary conditions $P_0 = 0$ and $P_N = 1$.
Rearranging (using $p + q = 1$):

$$p(P_{k+1} - P_k) = q(P_k - P_{k-1})$$

Let $D_k = P_{k+1} - P_k$. Then $D_k = (q/p)\, D_{k-1}$, so consecutive differences form a geometric sequence:

$$D_k = (q/p)^k\, D_0$$

Case 1: Biased game ($p \neq q$). Summing the telescoping differences:

$$P_k = P_0 + \sum_{i=0}^{k-1} D_i = D_0 \sum_{i=0}^{k-1} (q/p)^i = D_0 \cdot \frac{1 - (q/p)^k}{1 - q/p}$$

Applying $P_0 = 0$ (already satisfied) and $P_N = 1$:

$$1 = D_0 \cdot \frac{1 - (q/p)^N}{1 - q/p}$$

Solving for D_0 and substituting back:

$$P_k = \frac{1 - (q/p)^k}{1 - (q/p)^N}$$

Case 2: Fair game ($p = q = 1/2$). Now $q/p = 1$, so $D_k = D_0$ for all k — constant differences, meaning P_k is linear. With $P_0 = 0$ and $P_N = 1$:

$$P_k = \frac{k}{N}$$

This matches the $p \neq q$ formula in the limit as $q/p \to 1$ (by L'Hopital's rule).
Quant application: Survival probability of a trading strategy with limited capital. If $p < 1/2$ (negative edge), $(q/p)^k$ grows exponentially and $P_k \to 0$ for large N — ruin is nearly certain regardless of starting capital.

Problem 10: Two Envelopes

Problem
One envelope has X, other has $2X$. You pick one, see \$100. Should you switch?

Solution
The naive argument goes as follows: you see \$100, so the other envelope contains either \$50 or \$200, each with probability 1/2. The expected value of switching is:

$$\mathbb{E}[\text{switch}] = \frac{1}{2}(50) + \frac{1}{2}(200) = 125 > 100.$$

This suggests you should always switch — but by the same logic, you would switch back, leading to an infinite loop.
Resolution. The error is in assuming $P(\text{yours is the larger}) = P(\text{yours is the smaller}) = 1/2$ regardless of the observed amount. This implicitly requires a uniform prior over all positive reals, which is not a valid (proper) probability distribution.
With any proper prior on the amount X placed in the smaller envelope, observing \$100 updates your beliefs about whether you hold X or $2X$. Under any such prior, the expected gain from switching is exactly zero — the paradox disappears.

Problem 11: Expected Flips for HH

Problem
Flip fair coin until two consecutive Heads. Expected flips?

Hint
Use **first-step analysis**, but note that you need to track more than just "how many flips so far." The key state variable is what your last flip was — define states based on your current streak of consecutive heads (0 or 1).

- Let E be the expected flips from the start (streak of 0 heads).
- Let E_H be the expected additional flips given the last flip was H (streak of 1 head).
- From each state, condition on the next flip and set up a recurrence.

You'll get a small system of linear equations to solve.

Solution
Define two states based on the current streak of consecutive heads:

- E: expected flips from the start (streak of 0 heads).
- E_H: expected additional flips given the last flip was H (streak of 1 head).

From the start. Flip once. With probability $1/2$ it is T and we restart; with probability $1/2$ it is H and we move to state E_H:

$$E = 1 + \frac{1}{2}E + \frac{1}{2}E_H.$$

From state H. Flip once. With probability $1/2$ it is T and we restart from the beginning; with probability $1/2$ it is H and we are done (two consecutive heads):

$$E_H = 1 + \frac{1}{2}E.$$

Substituting the second equation into the first:

$$E = 1 + \frac{1}{2}E + \frac{1}{2}\left(1 + \frac{1}{2}E\right) = \frac{3}{2} + \frac{3}{4}E.$$

Solving:

$$\frac{1}{4}E = \frac{3}{2} \implies E = \mathbf{6}.$$

Problem 12: Fixed Points (Indicators)

Problem
A random permutation of $\{1, 2, \ldots, n\}$ is drawn. Expected number of fixed points (elements in their original position)?

Solution
Let X denote the total number of fixed points. Define indicator random variables $\mathbf{1}_i$ for each position i:

$$\mathbf{1}_i = \begin{cases} 1 & \text{if element } i \text{ is in position } i, \\ 0 & \text{otherwise.} \end{cases}$$

Then $X = \sum_{i=1}^{n} \mathbf{1}_i$. Each element is equally likely to land in any of the n positions, so:

$$P(\mathbf{1}_i = 1) = \frac{1}{n}.$$

By linearity of expectation (which holds regardless of dependence):

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[\mathbf{1}_i] = \sum_{i=1}^{n} \frac{1}{n} = \mathbf{1}.$$

The expected number of fixed points is exactly 1, regardless of n.

Problem 13: [J3.1] Die Payoff

Problem
A player tosses a fair die once. The payoff is \$1 for each dot on the upturned face. What is the fair ticket price?

Solution
Let X be the number of dots on the upturned face. Since each face is equally likely:

$$\mathbb{E}[X] = \frac{1}{6}(1+2+3+4+5+6) = \frac{21}{6} = \$3.50.$$

The fair ticket price is **\$3.50**.

Problem 14: [J3.2] Optimal Stopping

Problem
You roll a die up to three times. After each roll, you can take the number as dollars or roll again. What are your expected winnings?

Hint
Use **backward induction**. Start from the last die and work backwards:

1. At the last die, you must take whatever you roll, so $\mathbb{E}[\text{payoff}] = 3.5$.
2. At each earlier die, you take the roll if it exceeds the expected value of continuing — otherwise you roll again.
3. The threshold at each step is the expected value you computed for the remaining dice.

Compute $\mathbb{E}[\text{payoff from last die}]$ first, then use it to find the optimal strategy and expected value with two dice remaining, and so on.

Solution
Use **backward induction**. Let E_i denote the expected payoff with i rolls remaining under optimal play. The strategy at each step is: take the current roll if it exceeds the expected value of continuing; otherwise, roll again.
One roll remaining (E_1). You must accept whatever you roll:

$$E_1 = \frac{1}{6}(1+2+3+4+5+6) = 3.5.$$

Two rolls remaining (E_2). Take the roll if it exceeds $E_1 = 3.5$, i.e., take 4, 5, or 6 (probability 1/2); otherwise roll again:

$$E_2 = \frac{1}{2} \cdot \frac{4+5+6}{3} + \frac{1}{2} \cdot E_1 = \frac{1}{2}(5) + \frac{1}{2}(3.5) = 4.25.$$

Three rolls remaining (E_3). Take the roll if it exceeds $E_2 = 4.25$, i.e., take 5 or 6 (probability 1/3); otherwise roll again:

$$E_3 = \frac{1}{3} \cdot \frac{5+6}{2} + \frac{2}{3} \cdot E_2 = \frac{1}{3}(5.5) + \frac{2}{3}(4.25) = \frac{14}{3} \approx \mathbf{4.67}.$$

Problem 15: [J3.3] Box Game

Problem
Four sealed boxes: one contains \$100, three are empty. Pay X to open a box, repeat as desired. For a fair game, what is X?

Hint
Think about **symmetry**. You are opening boxes in a random order until you find the prize. The prize is equally likely to be in any of the n boxes, so it is equally likely to be the 1st box you open, the 2nd, ..., the nth.
The expected cost depends on the expected position of the prize in your opening sequence. What is $\mathbb{E}[\text{number of boxes opened}]$? Use this to set $\mathbb{E}[\text{cost}] = \mathbb{E}[\text{reward}]$ for a fair game.

Solution
A rational player opens boxes one at a time until finding the prize. By symmetry, the prize is equally likely to be in the 1st, 2nd, ..., or nth box opened. The expected number of boxes opened is:

$$\mathbb{E}[\text{boxes opened}] = \frac{1}{n}(1 + 2 + \cdots + n) = \frac{n+1}{2}.$$

For the game to be fair, the expected cost must equal the expected reward:

$$\frac{n+1}{2} \cdot X = P \qquad \Longrightarrow \qquad X = \frac{2P}{n+1}.$$

Plugging in $n = 4$ boxes and $P = \$100$:

$$X = \frac{2 \times 100}{5} = \mathbf{\$40}.$$

Problem 16: [J3.4] Adversarial Guessing

Problem
I pick a number n from 1 to 100. If you guess correctly, I pay you $\$n$, otherwise zero. How much would you pay to play?

Hint
This is a **game theory** problem, not a straightforward probability calculation. Think about **mixed strategies** — if your opponent knows your strategy, can they exploit it?

- If you use a deterministic strategy (always guess the same number), your adversary simply avoids that number and you win nothing.
- So you need a **randomized** strategy. The key insight is to find a distribution over guesses such that no matter what number your opponent picks, your expected payoff is the same.
- This is the **minimax equilibrium**: you choose probabilities p_j for each guess j so that the quantity $p_j \cdot j$ is constant across all j.

Solution
This is a **game theory** problem. If you use a deterministic strategy (always guess k), your adversary simply never picks k and you win nothing. You must **randomize**.
Let p_j be the probability you guess number j. If the adversary picks number j, your expected payout is $p_j \cdot j$. The adversary chooses j to minimize your payout, so at equilibrium you want $p_j \cdot j$ to be the same constant c for every j:

$$p_j \cdot j = c \quad \text{for all } j = 1, \ldots, 100 \qquad \Longrightarrow \qquad p_j = \frac{c}{j}.$$

Since the probabilities must sum to 1:

$$\sum_{j=1}^{100} p_j = c \sum_{j=1}^{100} \frac{1}{j} = c\, H_{100} = 1 \qquad \Longrightarrow \qquad c = \frac{1}{H_{100}}.$$

The harmonic number $H_{100} \approx 5.19$, so your guaranteed expected payoff is $c \approx \$0.19$. The fair price for this game is $\mathbf{1/H_{100} \approx \$0.19}$.

Problem 17: [J3.5] Doubling Game

Problem
You have a fair coin and start with \$1. If you toss H, your position doubles; if T, it halves. What is the expected value if you toss infinitely?

Solution
Let V_n denote the expected value of your position after n flips. After a single flip, Heads doubles your position and Tails halves it, so the expected multiplicative factor per flip is:

$$\mathbb{E}[\text{multiplier}] = \frac{1}{2}(2) + \frac{1}{2}\left(\frac{1}{2}\right) = \frac{5}{4}.$$

Since the flips are independent, the expected value after n flips is:

$$V_n = \left(\frac{5}{4}\right)^n.$$

As $n \to \infty$, $V_n \to \infty$. The expected value of the game is **infinite**.
Remark: Despite the infinite expected value, the *median* outcome shrinks to zero — almost surely the position converges to 0. This is because the log-growth rate is $\frac{1}{2}\ln 2 + \frac{1}{2}\ln(1/2) = 0$, so the typical path does not grow. The divergence of the expectation is driven by rare, extremely large outcomes.

Problem 18: [J3.6] Geometric Distribution

Problem
Toss a fair coin until you get a head. Let N be the number of tosses (including the final head). What is $\mathbb{E}(N)$ and $\mathrm{Var}(N)$?

Solution
Let N be the number of tosses. The probability of needing exactly i tosses is:

$$P(N = i) = \left(\frac{1}{2}\right)^i, \quad i = 1, 2, 3, \ldots$$

Computing $\mathbb{E}[N]$. We need to evaluate:

$$\mathbb{E}[N] = \sum_{i=1}^{\infty} i \cdot \frac{1}{2^i}$$

Start from the geometric series $\sum_{i=0}^{\infty} x^i = \frac{1}{1-x}$ and differentiate both sides with respect to x:

$$\sum_{i=1}^{\infty} i\, x^{i-1} = \frac{1}{(1-x)^2}$$

Multiplying both sides by x:

$$\sum_{i=1}^{\infty} i\, x^i = \frac{x}{(1-x)^2}$$

Setting $x = 1/2$:

$$\mathbb{E}[N] = \frac{1/2}{(1/2)^2} = 2$$

Computing $\mathrm{Var}(N)$. Differentiating the generating function once more and evaluating at $x = 1/2$ gives $\mathbb{E}[N^2] = 6$. Therefore:

$$\mathrm{Var}(N) = \mathbb{E}[N^2] - (\mathbb{E}[N])^2 = 6 - 4 = 2$$

Alternative (subtraction trick). Let $S = \sum_{i=1}^{\infty} i/2^i$. Then $S/2 = \sum_{i=1}^{\infty} i/2^{i+1}$. Subtracting term by term:

$$\frac{S}{2} = \sum_{i=1}^{\infty} \frac{1}{2^i} = 1$$

so $S = 2$.

Problem 19: [J3.7] HH or TT

Problem
Fair coin game: the game stops when either two heads (HH) or two tails (TT) appear consecutively. What is the expected number of tosses?

Solution
By symmetry the game behaves the same whether the last flip was H or T, so define a single state variable. Let E_1 be the expected number of additional flips to finish the game given that the most recent flip was either H or T.
From this state, we flip once. With probability $\frac{1}{2}$ the flip matches the previous one (game over), and with probability $\frac{1}{2}$ it differs (we remain in the same type of state, since we now have a single most-recent flip of the opposite face):

$$E_1 = 1 + \frac{1}{2}(0) + \frac{1}{2}(E_1)$$

Solving: $E_1 = 2$.
The game always begins with one toss to enter a state (the first flip can never end the game). So the total expected number of tosses is:

$$\mathbb{E}[\text{total}] = 1 + E_1 = 1 + 2 = 3$$

Problem 20: [J3.8] Three Consecutive Heads

Problem
For a fair coin, what is the expected number of tosses to get three heads in a row?

Hint
This is **first-step analysis** with multiple states. Define states based on your current streak of consecutive heads: 0, 1, or 2 heads in a row. Let E_i be the expected number of flips to reach 3 consecutive heads starting from state i.
From each state, consider what happens on the next flip:

- A heads extends the streak: state $i \to i + 1$.
- A tails resets the streak: state $i \to 0$.

This gives you three equations in three unknowns (E_0, E_1, E_2). Solve by substitution.

Solution
Let E_i be the expected number of additional flips needed to reach three consecutive heads, given that the current streak of consecutive heads is i. We want E_0.
Setting up the recurrences. From each state, we flip once. Heads extends the streak; tails resets it to zero.

- **State 2** (two heads in a row): heads completes the goal; tails resets to state 0.

$$E_2 = 1 + \frac{1}{2}(0) + \frac{1}{2}E_0 = 1 + \frac{1}{2}E_0$$

- **State 1** (one head in a row): heads advances to state 2; tails resets to state 0.

$$E_1 = 1 + \frac{1}{2}E_2 + \frac{1}{2}E_0$$

- **State 0** (no streak): heads advances to state 1; tails stays at state 0.

$$E_0 = 1 + \frac{1}{2}E_1 + \frac{1}{2}E_0$$

Solving. From the state-0 equation, rearrange:

$$\frac{1}{2}E_0 = 1 + \frac{1}{2}E_1 \quad \Longrightarrow \quad E_0 = 2 + E_1$$

Substitute the state-2 equation into the state-1 equation:

$$E_1 = 1 + \frac{1}{2}\left(1 + \frac{1}{2}E_0\right) + \frac{1}{2}E_0 = \frac{3}{2} + \frac{3}{4}E_0$$

Now substitute $E_1 = E_0 - 2$:

$$E_0 - 2 = \frac{3}{2} + \frac{3}{4}E_0 \quad \Longrightarrow \quad \frac{1}{4}E_0 = \frac{7}{2} \quad \Longrightarrow \quad E_0 = 14$$

Problem 21: 5 Consecutive Heads

Problem
Expected coin flips to get 5 consecutive heads?

Solution
Let E_k denote the expected number of additional flips needed to reach 5 consecutive heads, given that we currently have a streak of k consecutive heads. The boundary condition is $E_5 = 0$ (we are done).
For $k < 5$, condition on the next flip. With probability $\frac{1}{2}$ we flip heads and extend the streak to $k+1$; with probability $\frac{1}{2}$ we flip tails and the streak resets to zero:

$$E_k = 1 + \tfrac{1}{2}\, E_{k+1} + \tfrac{1}{2}\, E_0$$

Working backwards from E_4:

$$E_4 = 1 + \tfrac{1}{2}(0) + \tfrac{1}{2}\, E_0 = 1 + \tfrac{E_0}{2}$$

$$E_3 = 1 + \tfrac{1}{2}\, E_4 + \tfrac{1}{2}\, E_0 = 1 + \tfrac{1}{2}\left(1 + \tfrac{E_0}{2}\right) + \tfrac{E_0}{2} = \tfrac{3}{2} + \tfrac{3E_0}{4}$$

Continuing this pattern for E_2, E_1, and E_0 yields a single equation in E_0. Solving the system gives:

$$E_0 = 62$$

Pattern: For n consecutive heads with a fair coin, $E \approx 2^{n+1} - 2$.

Problem 22: [J3.9] Biased Coin

Problem
You toss a biased coin with $P(H) = p$. What is the expected number of tosses until a head? Until two consecutive heads?

Solution
Part 1: Expected tosses until the first head.
Let N be the number of tosses until the first head. The number of tails before the first head is geometric, so:

$$\mathbb{E}[N] = \sum_{i=1}^{\infty} i \cdot (1-p)^{i-1}\, p = \frac{1}{p}$$

Part 2: Expected tosses until two consecutive heads.
Let E_0 be the expected number of tosses from the start (no streak), and let E_1 be the expected number of additional tosses given that the most recent toss was heads.
From state 1, we flip once. With probability p we get another head (done), and with probability $1-p$ we get tails (reset to state 0):

$$E_1 = 1 + p \cdot 0 + (1-p)\, E_0 = 1 + (1-p)\, E_0$$

From state 0, we flip once. With probability p we advance to state 1, and with probability $1-p$ we remain at state 0:

$$E_0 = 1 + p\, E_1 + (1-p)\, E_0$$

Rearranging the second equation: $p\, E_0 = 1 + p\, E_1$, so $E_0 = \frac{1}{p} + E_1$. Substituting the expression for E_1:

$$E_0 = \frac{1}{p} + 1 + (1-p)\, E_0 \quad \Longrightarrow \quad p\, E_0 = \frac{1+p}{p}$$

$$E_0 = \frac{1+p}{p^2}$$

Summary: $\mathbb{E}[\text{first H}] = \dfrac{1}{p}, \quad \mathbb{E}[\text{two consecutive H}] = \dfrac{1+p}{p^2}.$

Problem 23: [J3.10] Double-Headed Coin

Problem
A bag contains 9 ordinary coins and 1 double-headed coin. You pick a coin at random and flip it 3 times, getting heads each time. What is the probability it is the double-headed coin?

Solution
We solve the general version with n total coins (one double-headed, $n-1$ fair) and k observed heads. Let D denote the event that the chosen coin is double-headed, and let H_k denote the event of observing k heads in k flips.
Prior probabilities. The coin is chosen uniformly at random:

$$P(D) = \frac{1}{n}, \qquad P(D^c) = \frac{n-1}{n}$$

Likelihoods. The double-headed coin always shows heads, while a fair coin shows k heads with probability $(1/2)^k$:

$$P(H_k \mid D) = 1, \qquad P(H_k \mid D^c) = \left(\frac{1}{2}\right)^k$$

Applying Bayes' theorem:

$$P(D \mid H_k) = \frac{P(H_k \mid D)\, P(D)}{P(H_k \mid D)\, P(D) + P(H_k \mid D^c)\, P(D^c)} = \frac{\frac{1}{n}}{\frac{1}{n} + \frac{1}{2^k} \cdot \frac{n-1}{n}} = \frac{2^k}{2^k + n - 1}$$

For the given values $n = 10$ and $k = 3$:

$$P(D \mid H_3) = \frac{8}{8+9} = \frac{8}{17}$$

Problem 24: [J3.11] Laplace's Rule

Problem
You take a coin from your pocket and flip it 3 times, getting heads each time. What is the probability the next flip is heads? What if you had flipped 100 heads?

Hint
The coin's bias p is unknown. Assume every value of p in $[0, 1]$ is equally likely a priori (flat prior). After observing data, use Bayes' theorem to update the density of p, then compute the expected value of p under that updated density.

Solution
If you *know* the coin is fair, then $P(\text{next H}) = 1/2$ regardless of past flips.
The interesting case is when the coin's bias is unknown. Let p denote the probability of heads. We assume a flat prior: every value of p in $[0, 1]$ is equally likely, so the prior density is $f(p) = 1$.
Updating after n heads in n flips. By Bayes' theorem, the posterior density of p is proportional to the likelihood times the prior:

$$f(p \mid n \text{ heads}) \propto p^n \cdot 1 = p^n$$

To normalize, we need $\int_0^1 p^n \, dp$. By the power rule this equals $\frac{1}{n+1}$, so the posterior density is:

$$f(p \mid n \text{ heads}) = (n+1)\, p^n$$

Predictive probability. The probability the next flip is heads equals the expected value of p under the posterior:

$$P(\text{next H} \mid n \text{ heads}) = \int_0^1 p \cdot (n+1)\, p^n \, dp = (n+1) \int_0^1 p^{n+1} \, dp$$

Evaluating the integral by the power rule again:

$$(n+1) \cdot \frac{1}{n+2} = \frac{n+1}{n+2}$$

This is **Laplace's rule of succession**. Plugging in:

- For $n = 3$: $P = 4/5 = 0.8$.
- For $n = 100$: $P = 101/102 \approx 0.99$.

Note: you may recognize the posterior $f(p) = (n+1)\, p^n$ as a Beta$(n+1, 1)$ distribution. The general version of this argument with k heads in n flips yields a Beta$(k+1, n-k+1)$ posterior, whose mean is $(k+1)/(n+2)$.

Problem 25: [J3.12] Pattern Counting

Problem
You flip a fair coin one million times. What is the expected number of occurrences of 6 heads followed by 6 tails?

Solution
Define indicator random variables: let $X_i = 1$ if the pattern HHHHHHTTTTTT begins at position i, and $X_i = 0$ otherwise.
The pattern has length 12, so valid starting positions are $i = 1, 2, \ldots, 10^6 - 11$. Each position has the same probability of matching:

$$P(X_i = 1) = \left(\frac{1}{2}\right)^{12} = \frac{1}{4096}$$

By **linearity of expectation** (which does not require independence of the X_i):

$$\mathbb{E}\left[\sum_{i=1}^{10^6-11} X_i\right] = (10^6 - 11) \cdot \frac{1}{4096} = \frac{999{,}989}{4096} \approx 244$$

Remark: the pattern HHHHHHTTTTTT cannot overlap with itself (it starts with H and ends with T), so the indicators happen to be pairwise independent at overlapping positions. However, linearity of expectation applies regardless.

Problem 26: [J3.13] Dart Board

Problem
You throw a dart at a circular board of radius R (assume uniform distribution). What is your expected distance from the center?
Follow-up: You win \$1 if you hit inside radius $R/2$ ten times in a row. You pay 10 cents per try. Over 100 tries, what is your expected profit/loss? What if you are a professional with $P(\text{inside } R/2) = 2 \cdot P(\text{outside } R/2)$?

Hint
Switch to **polar coordinates**. For a uniform point in the unit disk, the joint density in Cartesian is $f(x, y) = 1/(\pi R^2)$. In polar coordinates (r, θ), don't forget the Jacobian: $dA = r\,dr\,d\theta$.
You need the marginal distribution of r (the distance from center). The CDF is $P(r \leq t) = t^2/R^2$ for $t \in [0, R]$, which gives PDF $f(r) = 2r/R^2$. Then compute $\mathbb{E}[r]$ directly.

Solution
Uniform on disk means density $f = \frac{1}{\pi R^2}$. Using polar coordinates:

$$\mathbb{E}[r] = \int_0^{2\pi} \int_0^R r \cdot \frac{1}{\pi R^2} \cdot r \, dr \, d\theta = \frac{2}{R^2} \int_0^R r^2 \, dr = \frac{2}{R^2} \cdot \frac{R^3}{3} = \frac{2R}{3}$$

Follow-up. For a uniform dart, the probability of landing inside radius $R/2$ is the ratio of areas:

$$P(\text{inside } R/2) = \frac{\pi(R/2)^2}{\pi R^2} = \frac{1}{4}$$

The probability of hitting inside $R/2$ ten times in a row (assuming independent throws):

$$P(10 \text{ in a row}) = \left(\frac{1}{4}\right)^{10} = 2^{-20} \approx 10^{-6}$$

Over 100 tries, the expected number of wins is $100 \cdot 2^{-20} \approx 10^{-4}$, so the expected prize is approximately \$0.0001. The total cost is $\$0.10 \times 100 = \$10$. The expected loss is approximately \$10.
Professional. Here $P(\text{in}) = 2 \cdot P(\text{out}) = 2(1 - P(\text{in}))$, which gives $P(\text{in}) = \frac{2}{3}$. Then:

$$P(10 \text{ in a row}) = \left(\frac{2}{3}\right)^{10} \approx 0.017$$

The expected prize over 100 tries is $100 \times 0.017 \times \$1 \approx \$1.73$. The cost is still \$10, so the expected loss is approximately \$8.27.

Problem 27: [J3.15] Clarissa

Problem
A family has two children. You know at least one is a girl. What is $P(\text{both girls})$? What if you know the eldest is a girl? What if you know they have a girl named Clarissa?

Solution
List the sample space as (elder, younger): $\{BB, BG, GB, GG\}$, each with probability $1/4$.
(a) At least one is a girl. This eliminates BB, leaving $\{BG, GB, GG\}$. Only one of three outcomes has both girls:

$$P(\text{both girls} \mid \text{at least one girl}) = \frac{1}{3}$$

(b) The eldest is a girl. This restricts to $\{GB, GG\}$. Half of these have both girls:

$$P(\text{both girls} \mid \text{eldest is girl}) = \frac{1}{2}$$

(c) They have a girl named Clarissa. Knowing a specific name identifies a specific child, much like knowing "the eldest is a girl." The named child is fixed as a girl, and the other child is independent:

$$P(\text{both girls} \mid \text{a girl named Clarissa}) = \frac{1}{2}$$

Problem 28: [J3.16] Business Day

Problem
What is the probability that the fourth business day of the month is a Thursday?

Solution
Business days are Mon–Fri. Check each starting day:

1st is	4th BD date	Weekday
Mon	4	Thu ✓
Tue	4	Fri
Wed	6	Mon
Thu	6	Tue
Fri	6	Wed
Sat	6	Thu ✓
Sun	5	Thu ✓

Thursday when 1st is Mon, Sat, or Sun. $P = 3/7$.

Problem 29: [J3.17] Russian Roulette

Problem
A six-shooter revolver has two bullets in neighboring chambers. The barrel is spun, trigger pulled, and the gun does not fire. You are next—do you spin again or pull the trigger?

Solution
Label the six chambers 1 through 6 in order, with bullets in chambers 1 and 2 (adjacent).
Since the gun did not fire, the current chamber is one of $\{3, 4, 5, 6\}$, each equally likely.
Option 1: Spin again. A fresh spin selects uniformly among all 6 chambers, so:

$$P(\text{die} \mid \text{spin}) = \frac{2}{6} = \frac{1}{3}$$

Option 2: Pull the trigger (advance one chamber). The next chamber is 4, 5, 6, or 1, respectively. Only the transition $6 \to 1$ lands on a bullet:

$$P(\text{die} \mid \text{no spin}) = \frac{1}{4}$$

Since $1/4 < 1/3$, the better strategy is to **pull the trigger** without spinning.

Problem 30: [J3.20] Matching Socks

Problem
A drawer has 2 red socks and 2 black socks. You pull out 2 socks at random. What is the probability they match?

Solution
Draw first sock (any color). Of 3 remaining, exactly 1 matches. $P(\text{match}) = 1/3$.

Problem 31: [J3.22] Ant on Cube

Problem
An ant walks on the edges of a cube, taking 1 minute per edge. At each vertex, it randomly picks one of the 3 available edges. Starting from a vertex, what is the expected time to return to that vertex?

Hint
Think about this as a **random walk on a graph**. The cube graph has 8 vertices, each of degree 3 — it is a **regular graph**.
For any random walk on a finite, connected, undirected graph, the **stationary distribution** π and the **expected return time** are related by:

$$\mathbb{E}[\text{return time to } v] = \frac{1}{\pi(v)}.$$

For a regular graph (all vertices have the same degree), the stationary distribution is uniform. How many vertices does the cube have?

Solution
A cube has 8 vertices, each of degree 3. Because the graph is connected and regular (every vertex has the same degree), the stationary distribution of the random walk is uniform:

$$\pi(v) = \frac{1}{8} \quad \text{for every vertex } v$$

A standard result for random walks on finite, connected, undirected graphs is that the expected return time to any vertex v equals the reciprocal of its stationary probability:

$$\mathbb{E}[\text{return time to } v] = \frac{1}{\pi(v)} = 8 \text{ minutes}$$

Problem 32: [J3.39] Brownian Motion

Problem
What is Brownian motion?

Hint
Start from a symmetric random walk on a lattice with step size δx and time step δt. What relationship between δx and δt keeps the variance of the displacement finite and nonzero as both go to zero?

Solution
Construction from a random walk. Consider a symmetric random walk: at each time step δt, a particle moves $+\delta x$ or $-\delta x$ with equal probability. After $N = T/\delta t$ steps, the displacement is the sum of N iid random variables each with mean zero and variance δx^2. The total variance is:

$$\text{Var}[\text{displacement}] = \delta x^2 \cdot \frac{T}{\delta t}$$

For this to remain finite and nonzero as $\delta t \to 0$, we need $\delta x^2/\delta t$ to stay constant. This gives the scaling:

$$\delta x = \sqrt{\delta t}$$

The limit. Taking $\delta t \to 0$ with this scaling, the sum of N iid steps (each with variance δt) converges by the central limit theorem to a Gaussian with variance T. The limiting process W_t is called *standard Brownian motion* (or a Wiener process) and has three defining properties:

1. **Continuous paths**: W_t is continuous in t (with probability 1).
2. **Independent increments**: for any $0 \le s < t < u < v$, the increments $W_t - W_s$ and $W_v - W_u$ are independent.
3. **Gaussian increments**: $W_t - W_s \sim N(0, t-s)$.

Key consequence. The scaling $\delta x \sim \sqrt{\delta t}$ means Brownian paths are much rougher than smooth curves: displacement grows like $\sqrt{t}$, not t. In fact, Brownian motion is continuous everywhere but differentiable nowhere.

Problem 33: [J3.29] Order Statistics

Problem
Suppose $x_1, x_2, \ldots, x_n$ are i.i.d. uniform on $[0,1]$. What is $\mathbb{E}[\max] - \mathbb{E}[\min]$?

Solution
We use the tail-sum formula for the expectation of a non-negative random variable: $\mathbb{E}[X] = \int_0^\infty P(X > t)\,dt$.
Expected maximum. Let $M = \max(x_1, \ldots, x_n)$. Since each $x_i \sim U[0,1]$, we have $P(M \le t) = t^n$ for $t \in [0,1]$, so

$$P(M > t) = 1 - t^n.$$

Therefore

$$\mathbb{E}[M] = \int_0^1 (1 - t^n)\,dt = 1 - \frac{1}{n+1} = \frac{n}{n+1}.$$

Expected minimum. Let $m = \min(x_1, \ldots, x_n)$. All n variables must exceed t for the minimum to exceed t:

$$P(m > t) = (1-t)^n.$$

Therefore

$$\mathbb{E}[m] = \int_0^1 (1-t)^n\,dt = \frac{1}{n+1}.$$

Answer.

$$\mathbb{E}[M] - \mathbb{E}[m] = \frac{n}{n+1} - \frac{1}{n+1} = \frac{n-1}{n+1}.$$

Problem 34: [J3.28] Uniform Distribution

Problem
What is the distribution function of a uniform $U(a, b)$ random variable?

Solution
The density is $f(x) = \frac{1}{b-a}$ for $x \in [a, b]$.
The CDF is:

$$F(x) = \int_a^x \frac{1}{b-a}\, dt = \frac{x-a}{b-a} \quad \text{for } x \in [a, b]$$

with $F(x) = 0$ for $x < a$ and $F(x) = 1$ for $x > b$.

Problem 35: [J3.31] Density Transformation

Problem
A random variable X has density $f_X(x)$. What is the density of $g(X)$?

Solution Needed
This problem is awaiting a solution. See `UNSOLVED.md` for details.

Problem 36: [J3.33] Convolution: Sum of Uniforms

Problem
Let X and Y be independent uniform on $[0, 1]$. What is the density of $X + Y$?

Hint
Since X and Y are independent, $f(x, y) = 1$ on the unit square. Find $P(X + Y \leq s)$ by computing the area of $\{(x, y) \in [0, 1]^2 : x + y \leq s\}$, then differentiate. The geometry changes at $s = 1$.

Solution
Let $S = X + Y$. Since $X, Y \in [0,1]$, the sum S ranges from 0 to 2. By independence, the joint density is $f(x,y) = 1$ on the unit square, so $P(X+Y \le s)$ equals the area of the region $\{(x,y) \in [0,1]^2 : x + y \le s\}$.
Case $0 \le s \le 1$**.** The line $x + y = s$ has intercepts $(s,0)$ and $(0,s)$, both inside the unit square. The region $x + y \le s$ is a triangle in the bottom-left corner with legs of length s:

$$P(X+Y \le s) = \frac{s^2}{2}$$

Case $1 < s \le 2$**.** The line $x + y = s$ now cuts off a triangle in the top-right corner where $x + y > s$. That triangle has legs of length $2 - s$, so:

$$P(X+Y \le s) = 1 - \frac{(2-s)^2}{2}$$

Density. Differentiating $P(X+Y \le s)$ with respect to s:

$$f_S(s) = \begin{cases} s & 0 \le s \le 1 \\ 2 - s & 1 < s \le 2 \end{cases}$$

This is a triangular density peaking at $s = 1$, which matches the intuition that values near the middle of $[0,2]$ are most likely.
Alternative: convolution. The density of a sum of independent random variables is the convolution of their densities. Since both densities are simply 1 on $[0,1]$, the convolution at s is just the length of the overlap between the intervals $[0,1]$ and $[s-1, s]$:

$$f_S(s) = \min(1, s) - \max(0, s-1)$$

This gives s when $s \le 1$ and $2 - s$ when $s > 1$, confirming the same triangular density.

Problem 37: [J3.30] Infinite Variance

Problem
Give an example of a distribution with infinite variance.

Solution
The standard example is the **Cauchy distribution**, with density

$$f(x) = \frac{1}{\pi(1+x^2)}, \qquad x \in \mathbb{R}.$$

The mean is undefined because the integral $\int x\, f(x)\, dx$ does not converge absolutely, and the variance is likewise infinite.
The underlying reason is that the tails decay as $1/x^2$, which is too slow for the second moment $\int x^2 f(x)\, dx$ to be finite.
Quant application: fat-tailed return distributions may have undefined higher moments, which breaks naive mean-variance optimization.

2 Statistics

Problem 1: Unbiased Variance Estimator

Problem
Why divide by $n-1$?

Solution
Intuition. The sample mean $\bar{x}$ minimizes $\sum(x_i - c)^2$ over all constants c. Because $\bar{x}$ is fit to the data, the sum of squared deviations around it is systematically smaller than the sum around the true mean μ. Dividing by n therefore underestimates the population variance; dividing by $n-1$ corrects for this.
Derivation. Consider a single term $\mathbb{E}[(X_i - \bar{X})^2]$. Expand the variance of the difference:

$$\mathrm{Var}(X_i - \bar{X}) = \mathrm{Var}(X_i) + \mathrm{Var}(\bar{X}) - 2\mathrm{Cov}(X_i, \bar{X}).$$

Since the X_i are i.i.d. with variance σ^2, we have $\mathrm{Var}(\bar{X}) = \sigma^2/n$ and $\mathrm{Cov}(X_i, \bar{X}) = \sigma^2/n$. Substituting:

$$\mathbb{E}[(X_i - \bar{X})^2] = \sigma^2 + \frac{\sigma^2}{n} - \frac{2\sigma^2}{n} = \sigma^2\left(1 - \frac{1}{n}\right).$$

Summing over all n terms:

$$\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \bar{X})^2\right] = (n-1)\,\sigma^2.$$

Dividing by $n-1$ yields an unbiased estimator:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2, \qquad \mathbb{E}[s^2] = \sigma^2.$$

Numerical example. Let $X_1, X_2, X_3 \sim N(10, 4)$ and suppose the observed samples are $\{8, 10, 12\}$. The sample mean is $\bar{x} = (8+10+12)/3 = 10$.

- *Biased estimator* (dividing by $n = 3$): $\hat{\sigma}_n^2 = \frac{(8-10)^2+(10-10)^2+(12-10)^2}{3} = \frac{8}{3} \approx 2.67$.
- *Unbiased estimator* (dividing by $n-1 = 2$): $s^2 = \frac{8}{2} = 4$, which equals the true variance $\sigma^2 = 4$.

The biased version underestimates by the factor $(n-1)/n = 2/3$.

Problem 2: Max of Uniform

Problem
$X_1, \ldots, X_n \sim U[0, \theta]$. MLE for θ?

Solution
Finding the MLE. The likelihood is

$$L(\theta) = \prod_{i=1}^{n} \frac{1}{\theta} \mathbf{1}_{0 \le x_i \le \theta} = \theta^{-n} \mathbf{1}\{\max(x_i) \le \theta\}.$$

For $\theta \ge \max(x_i)$ this equals θ^{-n}, which is strictly decreasing in θ. For $\theta < \max(x_i)$ the likelihood is zero. Therefore the maximum is attained at the smallest admissible value:

$$\hat{\theta}_{\text{MLE}} = \max(x_i).$$

Bias of the MLE. Let $Y = \max(X_1, \ldots, X_n)$. Its CDF and density are

$$P(Y \le y) = \left(\frac{y}{\theta}\right)^n, \qquad f_Y(y) = \frac{n\, y^{n-1}}{\theta^n}, \qquad 0 \le y \le \theta.$$

Computing the expectation:

$$\mathbb{E}[Y] = \int_0^\theta y \cdot \frac{n\, y^{n-1}}{\theta^n}\, dy = \frac{n}{\theta^n} \int_0^\theta y^n\, dy = \frac{n}{\theta^n} \cdot \frac{\theta^{n+1}}{n+1} = \frac{n}{n+1}\, \theta.$$

The MLE is biased downward. An unbiased estimator is $\dfrac{n+1}{n} \max(x_i)$.

Problem 3: Law of Large Numbers vs CLT

Problem
Difference?

Solution

- **LLN**: Convergence in Probability. The sample mean approaches the true mean as $n \to \infty$ (it becomes a constant). $\bar{X}_n \xrightarrow{P} \mu$.
- **CLT**: Convergence in Distribution. The *fluctuations* of the sample mean around the true mean, scaled by $\sqrt{n}$, approach a Normal distribution. $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$.

Problem 4: Confidence Interval Width

Problem
How does sample size affect CI width? How to halve the CI?

Solution
A Normal-based confidence interval takes the form

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

The total width is therefore

$$W = 2\, z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \propto \frac{1}{\sqrt{n}}.$$

To halve the width, we need $\sqrt{n}$ to double, i.e. n must increase to $4n$. **You must quadruple the sample size to cut the confidence interval width in half.**
Quant application: to halve the standard error on a Sharpe ratio estimate, you need a backtest four times as long.

Problem 5: Type I vs Type II Error

Problem
Define and explain the tradeoff.

Solution

- **Type I error (α)**: a false positive — rejecting H_0 when it is true. "Crying wolf."
- **Type II error (β)**: a false negative — failing to reject H_0 when it is false. "Missing the signal."
- **Power** $= 1 - \beta$: the probability of correctly rejecting a false H_0.

Tradeoff. Lowering α (a stricter threshold) increases β (more real effects are missed), and vice versa.
Quant application: setting a strict significance level for trading signals means fewer false discoveries but potentially missing real predictors.

Problem 6: Sufficient Statistics

Problem
What does it mean for a statistic to be sufficient?

Solution
A statistic $T(X)$ is **sufficient** for a parameter θ if the conditional distribution of the data X given $T(X)$ does not depend on θ.
Intuitively, T captures *all* information about θ contained in the data. Once you know T, no other function of X can improve your estimate of θ.
Factorization Theorem. T is sufficient for θ if and only if the likelihood factors as

$$L(\theta; x) = g\big(T(x), \theta\big) \cdot h(x),$$

where g depends on the data only through $T(x)$, and h depends only on the data (not on θ).
Example (Bernoulli). Let $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$. The sufficient statistic is $T = \sum_{i=1}^{n} X_i$, the total number of successes. The particular order in which heads and tails appear is irrelevant for estimating p.

Problem 7: Cramér-Rao Lower Bound

Problem
What's the minimum variance an unbiased estimator can achieve?

Solution
For any unbiased estimator $\hat{\theta}$ of θ, the Cramér–Rao inequality states

$$\mathrm{Var}(\hat{\theta}) \;\geq\; \frac{1}{I(\theta)},$$

where $I(\theta)$ is the **Fisher information**:

$$I(\theta) = \mathbb{E}\left[\left(\frac{\partial \ln f(X;\,\theta)}{\partial \theta}\right)^2\right].$$

Fisher information measures how "peaked" the likelihood is around the true value of θ. Greater curvature means the data are more informative about θ, so a lower estimation variance is achievable.
An estimator whose variance equals the Cramér–Rao bound is called **efficient**. For example, the sample mean is an efficient estimator of the mean of a Normal distribution.

Problem 8: [J3.34] Central Limit Theorem

Problem
What does the Central Limit Theorem say?

Solution
Let $X_1, \ldots, X_n$ be i.i.d. random variables with mean μ and finite variance σ^2. The Central Limit Theorem states that the standardized sum converges in distribution to a standard Normal:

$$\frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}} \;\xrightarrow{d}\; N(0,1) \quad \text{as } n \to \infty.$$

Equivalently, in terms of the sample mean $\bar{X}_n$:

$$\sqrt{n}\,(\bar{X}_n - \mu) \;\xrightarrow{d}\; N(0,\,\sigma^2).$$

Intuition: averages of anything (with finite variance) become approximately Normal. This is the theoretical foundation for z-tests and Normal-based confidence intervals.

Problem 9: [J3.32] kth Order Statistic

Problem
What is the distribution function and density function of the kth order statistic?

Solution Needed
This problem is awaiting a solution. See `UNSOLVED.md` for details.

3 Linear Algebra

Problem 1: Eigenvalues of Probability/Stochastic Matrix

Problem
If columns sum to 1 (Markov), show $\lambda_{max} = 1$.

Solution
Let $\mathbf{1} = [1, 1, \ldots, 1]^T$ denote the all-ones vector. Since the columns of A each sum to 1, we have $\sum_i A_{ij} = 1$ for every column j. Writing this in matrix form:

$$A^T \mathbf{1} = \mathbf{1}.$$

This shows that $\mathbf{1}$ is an eigenvector of A^T with eigenvalue 1. Since A and A^T share the same characteristic polynomial, they have the same eigenvalues. Therefore 1 is also an eigenvalue of A.
It remains to show that no eigenvalue exceeds 1 in magnitude. By the Perron–Frobenius theorem (applied to nonnegative matrices), the spectral radius is bounded by the maximum column sum. Since every column sums to exactly 1, the spectral radius satisfies

$$\rho(A) \leq 1.$$

Hence $\lambda_{\max} = 1$.

Problem 2: Rank One Update

Problem
If A is invertible, what is $(A + uv^T)^{-1}$?

Solution
The answer is given by the **Sherman–Morrison formula**:

$$(A + uv^T)^{-1} = A^{-1} - \frac{A^{-1}uv^TA^{-1}}{1 + v^TA^{-1}u},$$

provided the scalar $1 + v^TA^{-1}u \neq 0$.
Verification. Define $s = v^TA^{-1}u$, which is a scalar. Multiply the proposed inverse on the left by $(A + uv^T)$:

$$(A + uv^T)\left(A^{-1} - \frac{A^{-1}uv^TA^{-1}}{1 + s}\right).$$

Distributing the two terms:

$$= I + uv^TA^{-1} - \frac{uv^TA^{-1}}{1 + s} - \frac{uv^TA^{-1}u\,v^TA^{-1}}{1 + s}.$$

In the last term, $v^TA^{-1}u = s$, so this simplifies to

$$= I + uv^TA^{-1} - \frac{uv^TA^{-1}(1 + s)}{1 + s} = I + uv^TA^{-1} - uv^TA^{-1} = I.$$

Problem 3: [J3.26] SPD Correlation

Problem
Assets A, B, C have pairwise correlations $\rho_{AB} = 0.9$ and $\rho_{BC} = 0.9$. What is the valid range for ρ_{AC}?

Solution
The 3×3 correlation matrix

$$\Sigma = \begin{pmatrix} 1 & \rho_{AB} & \rho_{AC} \\ \rho_{AB} & 1 & \rho_{BC} \\ \rho_{AC} & \rho_{BC} & 1 \end{pmatrix}$$

must be positive semi-definite (PSD). A necessary condition is $\det(\Sigma) \geq 0$.
Deriving the constraint. Expanding the determinant and requiring it to be nonnegative yields a bound on ρ_{AC} in terms of the other two correlations:

$$\rho_{AC} \in \left[\rho_{AB}\rho_{BC} - \sqrt{(1-\rho_{AB}^2)(1-\rho_{BC}^2)},\ \ \rho_{AB}\rho_{BC} + \sqrt{(1-\rho_{AB}^2)(1-\rho_{BC}^2)}\right].$$

Geometric intuition: Correlations can be viewed as cosines of angles between unit vectors. If $\rho = \cos\theta$, then the formula above is exactly the cosine addition identity $\cos(\alpha \pm \beta)$.
Plugging in $\rho_{AB} = \rho_{BC} = 0.9$:

$$\rho_{AB}\rho_{BC} = 0.81, \qquad \sqrt{(1-0.81)(1-0.81)} = \sqrt{0.19 \times 0.19} = 0.19.$$

Therefore the valid range is

$$\rho_{AC} \in [0.81 - 0.19,\ 0.81 + 0.19] = [0.62,\ 1.0].$$

Problem 4: Eigenvalues of A^2 vs A

Problem
Relationship?

Solution
Let v be an eigenvector of A with eigenvalue λ.

$$A^2 v = A(Av) = A(\lambda v) = \lambda(Av) = \lambda(\lambda v) = \lambda^2 v.$$

The eigenvalues are squared. Eigenvectors remain the same.

Problem 5: Trace and Determinant

Problem
A 2x2 matrix has Trace 5 and Det 6. Eigenvalues?

Solution
Characteristic polynomial $\lambda^2 - \mathrm{tr}(A)\lambda + \det(A) = 0$.

$$\lambda^2 - 5\lambda + 6 = 0 \implies (\lambda - 2)(\lambda - 3) = 0.$$

Eigenvalues are $\{2, 3\}$.

Problem 6: Orthogonal Projection

Problem
Project vector b onto column space of matrix A. Formula?

Solution
We seek the vector $\hat{b}$ in the column space of A that is closest to b. Equivalently, we want to find $\hat{x}$ minimizing $\|b - Ax\|^2$.
Setting the gradient to zero gives the **normal equations**. The residual $b - A\hat{x}$ must be orthogonal to every column of A:

$$A^T(b - A\hat{x}) = 0.$$

Expanding and solving for $\hat{x}$:

$$A^T A\,\hat{x} = A^T b \implies \hat{x} = (A^T A)^{-1} A^T b.$$

The projected vector is therefore

$$\hat{b} = A\hat{x} = A(A^T A)^{-1} A^T b.$$

The matrix $P = A(A^T A)^{-1} A^T$ is called the **projection matrix**. It has two key properties:

- **Idempotent**: $P^2 = P$ (projecting twice is the same as projecting once).
- **Symmetric**: $P^T = P$ (the projection is orthogonal).

Quant application: This is exactly OLS regression — the fitted values $\hat{y} = Pb$ are the projection of the response vector onto the column space of the regressors.

Problem 7: Diagonalizability

Problem
When is an $n \times n$ matrix diagonalizable?

Solution
An $n \times n$ matrix A is diagonalizable if and only if it has n linearly independent eigenvectors. Equivalently, $A = PDP^{-1}$ where D is diagonal and the columns of P are eigenvectors.
Two important sufficient conditions:

- A has n *distinct* eigenvalues. In this case each eigenspace is one-dimensional, so the n eigenvectors are automatically linearly independent.
- A is symmetric (or Hermitian). The Spectral Theorem guarantees a full orthonormal eigenbasis.

Failure case: The matrix

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

has eigenvalue $\lambda = 0$ with algebraic multiplicity 2 but geometric multiplicity 1 (only one linearly independent eigenvector). Since we cannot find 2 independent eigenvectors, A is not diagonalizable.

Problem 8: Nilpotent Matrices

Problem
If $A^k = 0$ for some k, what are A's eigenvalues?

Solution
Let λ be an eigenvalue of A with corresponding eigenvector $v \neq 0$. By definition,

$$Av = \lambda v.$$

Applying A repeatedly, we get

$$A^k v = \lambda^k v.$$

Since $A^k = 0$, the left-hand side vanishes:

$$\lambda^k v = 0.$$

Because $v \neq 0$, we must have $\lambda^k = 0$, which forces

$$\lambda = 0.$$

Therefore every eigenvalue of a nilpotent matrix is zero. Two immediate consequences follow: the trace (sum of eigenvalues) satisfies $\mathrm{tr}(A) = 0$, and the determinant (product of eigenvalues) satisfies $\det(A) = 0$.
Example: Any strictly upper triangular matrix (zeros on and below the diagonal) is nilpotent.

Problem 9: [J3.35] Covariance Matrix is PSD

Problem
Prove that a covariance matrix is positive semi-definite.

Solution Needed
This problem is awaiting a solution. See `UNSOLVED.md` *for details.*

4 Regression

Problem 1: Univariate Coefficient

Problem
Formula for β in simple regression?

Solution

$$\hat{\beta} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)} = \rho_{XY}\frac{\sigma_Y}{\sigma_X}.$$

Intuition: It is the correlation scaled by the ratio of volatilities. If X moves 1σ, Y moves $\rho\sigma_Y$.

Problem 2: Adding Variables

Problem
What happens to R^2 when you add a variable? Adjusted R^2?

Solution
R^2 **always increases.** Recall that R^2 is the fraction of variance in Y explained by the regressors:

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}.$$

Adding a regressor expands the column space of X, so the projection of Y onto this larger subspace can only get closer (or stay the same). The residual sum of squares SS_{res} cannot increase, and therefore R^2 cannot decrease.
Adjusted R^2 can decrease. Adjusted R^2 penalizes model complexity:

$$R^2_{\text{adj}} = 1 - \frac{SS_{\text{res}}/(n-p-1)}{SS_{\text{tot}}/(n-1)},$$

where n is the sample size and p is the number of regressors. Adding a variable increases p, which increases the denominator penalty $n - p - 1$. If the new variable's t-statistic is less than roughly 1, the penalty outweighs the reduction in SS_{res}, and adjusted R^2 decreases—indicating the variable adds noise rather than signal.

Problem 3: Omitted Variable Bias

Problem
True model $Y = X_1\beta_1 + X_2\beta_2 + \epsilon$. You fit $Y = X_1\hat{\beta}_1 + \nu$. What is $\mathbb{E}[\hat{\beta}_1]$?

Solution

$$\begin{aligned}\hat{\beta}_1 &= (X_1^T X_1)^{-1} X_1^T Y = (X_1^T X_1)^{-1} X_1^T (X_1\beta_1 + X_2\beta_2 + \epsilon) \\ &= \beta_1 + (X_1^T X_1)^{-1} X_1^T X_2 \beta_2 + (X_1^T X_1)^{-1} X_1^T \epsilon.\end{aligned}$$

Taking expectations ($\mathbb{E}[\epsilon] = 0$):

$$\mathbb{E}[\hat{\beta}_1] = \beta_1 + \delta_{21}\beta_2$$

where δ_{21} is the regression coefficient of X_2 on X_1. If X_1, X_2 are uncorrelated, no bias.

Problem 4: Heteroskedasticity

Problem
What if $\mathrm{Var}(\epsilon_i) = \sigma_i^2$?

Solution
OLS relies on homoskedasticity for the Gauss-Markov theorem (BLUE). If variance is not constant:

- $\hat{\beta}$ remains **unbiased** and **consistent**.
- $\hat{\beta}$ is **inefficient** (variance is not minimal).
- Standard errors estimates are **wrong** (biased), invalidating t-tests.

Fix: Use Robust Standard Errors (White/Newey-West) or Weighted Least Squares (scale variables by $1/\sigma_i$).

Problem 5: Ridge vs Lasso

Problem
Penalty terms?

Solution
Regularization prevents overfitting by adding a penalty term that shrinks coefficient magnitudes. The two standard approaches differ in the norm used for the penalty.
Ridge regression (L_2 penalty). Ridge minimizes:

$$\hat{\beta}_{\text{ridge}} = \arg\min_{\beta} \left\{ \sum_{i=1}^{n} (y_i - x_i^T \beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right\}.$$

The closed-form solution is $\hat{\beta}_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T Y$. The λI term ensures invertibility even under multicollinearity. Ridge shrinks all coefficients toward zero but never sets any exactly to zero.
Lasso regression (L_1 penalty). Lasso minimizes:

$$\hat{\beta}_{\text{lasso}} = \arg\min_{\beta} \left\{ \sum_{i=1}^{n} (y_i - x_i^T \beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}.$$

Unlike Ridge, Lasso promotes **sparsity**: it drives some coefficients exactly to zero, thereby performing automatic feature selection. There is no closed-form solution; Lasso requires iterative algorithms (e.g., coordinate descent).
Geometric intuition. The L_2 constraint region is a sphere (smooth, no corners), while the L_1 constraint region is a diamond (sharp corners that lie on the axes). The OLS solution's contour ellipses are more likely to intersect the diamond at a corner, setting one or more coefficients to zero.

Problem 6: Multicollinearity

Problem
What happens when two regressors are highly correlated ($X_1 \approx X_2$)?

Solution
The matrix $X^T X$ becomes nearly singular (ill-conditioned).

- **Coefficients** $\hat{\beta}$ become *unstable*: small changes in data cause wild swings in estimates.
- **Standard Errors** inflate dramatically (wide confidence intervals).
- **Prediction** may still be fine—the model works as a black box, but individual β_j are uninterpretable.

Detection: A Variance Inflation Factor (VIF) > 10 suggests problematic multicollinearity.
Fixes: Drop redundant variables, use PCA to orthogonalize regressors, or apply Ridge regression (which stabilizes $X^T X$ by adding λI).

Problem 7: Durbin-Watson Test

Problem
How do you test for autocorrelation in residuals?

Solution
The Durbin-Watson statistic:

$$DW = \frac{\sum_{t=2}^{T}(\hat{e}_t - \hat{e}_{t-1})^2}{\sum_{t=1}^{T}\hat{e}_t^2} \approx 2(1-\hat{\rho})$$

where $\hat{\rho}$ is the first-order autocorrelation of residuals.

- $DW \approx 2$: No autocorrelation ($\hat{\rho} \approx 0$).
- $DW \approx 0$: Positive autocorrelation ($\hat{\rho} \approx 1$). Residuals trend together.
- $DW \approx 4$: Negative autocorrelation ($\hat{\rho} \approx -1$). Residuals oscillate.

Quant Application: Time series returns often exhibit autocorrelation. Ignoring it biases standard errors (Newey-West correction needed).

Problem 8: Logistic Regression

Problem
Why not use OLS for binary outcomes $Y \in \{0,1\}$?

Solution
Why not OLS? With a binary outcome $Y \in \{0,1\}$, OLS predicts $\hat{Y} = X\beta$, which can fall below 0 or above 1—nonsensical as a probability.
Logistic regression model. Instead, we model the conditional probability through the logistic (sigmoid) function:

$$P(Y=1 \mid X) = \frac{1}{1+e^{-X\beta}} = \sigma(X\beta).$$

Equivalently, the *log-odds* (logit) is linear in the predictors:

$$\ln \frac{P(Y=1 \mid X)}{1-P(Y=1 \mid X)} = X\beta.$$

Interpretation. Each coefficient β_j represents the change in log-odds per unit change in X_j. The corresponding *odds ratio* is e^{β_j}: a one-unit increase in X_j multiplies the odds of $Y=1$ by e^{β_j}.
Quant Application: Logistic regression is widely used for predicting binary events such as default/no-default or up/down day.

5 Time Series

Problem 1: AR(1) Half-Life

Problem
$X_t = 0.8X_{t-1} + \epsilon_t$. What is the half-life?

Solution
In an AR(1) process $X_t = \phi X_{t-1} + \varepsilon_t$, a unit shock decays geometrically: after h steps the impulse response is ϕ^h. The half-life is the number of steps until the response falls to half its initial value:

$$\phi^h = 0.5 \quad \Longrightarrow \quad h = \frac{\ln(0.5)}{\ln(\phi)}$$

Substituting $\phi = 0.8$:

$$h = \frac{\ln(0.5)}{\ln(0.8)} = \frac{-0.693}{-0.223} \approx 3.1 \text{ steps}$$

Intuition: A shock loses half its magnitude in about 3 periods. This is a key parameter for mean-reversion strategies: it tells you how quickly a dislocation corrects itself and therefore how long you should expect to hold a position.

Problem 2: Cointegration Check

Problem
You regress Y on X, residual ADF $t = -3.2$ vs crit -3.0 (5%). Is the pair cointegrated?

Solution
The Engle–Granger procedure tests for cointegration by checking whether the residuals from a levels regression are stationary:

1. Regress Y_t on X_t to get residuals $\hat{e}_t = Y_t - \hat{\beta} X_t$.
2. Run the Augmented Dickey–Fuller (ADF) test on $\hat{e}_t$.
3. Compare the ADF t-statistic to the critical value.

Here the ADF t-statistic is -3.2, which is more negative than the 5% critical value of -3.0. We therefore **reject the null hypothesis of a unit root** in the residuals.
This means the spread $Y_t - \hat{\beta} X_t$ is stationary: the pair is cointegrated, and deviations from the long-run relationship are mean-reverting.
Intuition: A stationary residual implies the spread has a restoring force. When the spread widens, it tends to revert — exactly the property a pairs-trading strategy requires.

Problem 3: GARCH Persistence

Problem
$\alpha = 0.05$, $\beta = 0.92$. What is the persistence? Is it stable?

Solution
In a GARCH(1,1) model, the conditional variance evolves as:

$$\sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2$$

The **persistence** is defined as $\alpha + \beta$:

$$\alpha + \beta = 0.05 + 0.92 = 0.97$$

This means that after a volatility shock, 97% of the shock persists into the next period — decay is very slow. The process is covariance-stationary because $\alpha + \beta < 1$. The long-run (unconditional) variance is:

$$\bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta}$$

Intuition: High persistence (≈ 1) means that volatility clustering is pronounced — periods of high volatility are followed by more high volatility, and calm periods persist. This is a universal stylized fact of financial returns.

Problem 4: One-Step AR(1) Forecast

Problem
$\hat{\phi} = 0.2$, $X_t = 10$ bps. What is $\hat{X}_{t+1}$?

Solution
The AR(1) model (with zero unconditional mean) is $X_{t+1} = \phi\,X_t + \varepsilon_{t+1}$. The one-step-ahead forecast is the conditional expectation:

$$\hat{X}_{t+1} = \hat{\phi} \cdot X_t = 0.2 \times 10 = 2 \text{ bps}$$

The signal shrinks from 10 bps to 2 bps in a single step.
Intuition: A small $\hat{\phi}$ means weak serial dependence — the process has short memory and reverts quickly toward its unconditional mean (here, zero). Most of today's value is noise that will not persist.

Problem 5: Ljung–Box Residual Check

Problem
For 20 lags, Q exceeds chi-square critical. What does this mean?

Solution
The Ljung–Box test checks whether the first m autocorrelations of a series are jointly zero. Under the null hypothesis (white noise), the test statistic Q follows a χ^2 distribution:

$$Q = N(N+2)\sum_{k=1}^{m}\frac{\hat{\rho}_k^2}{N-k} \;\sim\; \chi^2(m)$$

If Q exceeds the χ^2 critical value (here at 20 lags), we reject the null. This means the **residuals are not white noise** — significant autocorrelation remains unexplained by the model.
Action: The model is mis-specified. Consider adding more AR or MA lags, including seasonal terms, or changing the model structure entirely to capture the remaining serial dependence.
Intuition: Always check residuals for leftover autocorrelation before using a model for trading. Unexplained serial structure means the model is leaving predictable variation on the table — or worse, its forecasts are systematically biased.

6 Finance

Problem 1: Portfolio Opt (CAPM β)

Problem
What is CAPM β?

Solution
Definition. Let r_i be the return on asset i, r_M the return on the market portfolio, and r_f the risk-free rate. The CAPM beta of asset i is:

$$\beta_i = \frac{\mathrm{Cov}(r_i, r_M)}{\mathrm{Var}(r_M)}.$$

Interpretation. Beta measures the sensitivity of asset i's return to the market. Equivalently, it is the slope coefficient from regressing r_i on r_M:

$$r_i = \alpha_i + \beta_i r_M + \epsilon_i.$$

A stock with $\beta_i = 1.5$ is expected to move 1.5% for every 1% move in the market.
CAPM pricing equation. Under the Capital Asset Pricing Model, the expected return on asset i is linear in its beta:

$$\mathbb{E}[r_i] = r_f + \beta_i \left(\mathbb{E}[r_M] - r_f\right).$$

The term $\mathbb{E}[r_M] - r_f$ is the *equity risk premium*. The key insight is that only **systematic risk** (captured by β_i) is compensated with higher expected return. Idiosyncratic risk can be diversified away and therefore earns no premium.

Problem 2: Sharpe Ratio

Problem
Geometric interpretation?

Solution
Definition. Let r_p be the portfolio return, r_f the risk-free rate, and σ_p the portfolio volatility. The Sharpe ratio is:

$$SR = \frac{\mathbb{E}[r_p - r_f]}{\sigma_p}.$$

Geometric interpretation. In mean-volatility space (σ, μ), the risk-free asset sits at the point $(0, r_f)$. Any portfolio p sits at (σ_p, μ_p). The Sharpe ratio is the *slope* of the line (the Capital Allocation Line) connecting the risk-free asset to p.
Maximizing the Sharpe ratio therefore means finding the steepest possible line from $(0, r_f)$ to any feasible portfolio—this tangent portfolio is the mean-variance optimal risky portfolio.
Riemannian view. In the Riemannian manifold of assets, the Sharpe ratio corresponds to the cosine of the angle between the portfolio weight vector and the "skill" vector (alpha). Maximizing Sharpe is equivalent to aligning the portfolio as closely as possible with the direction of skill.

Problem 3: Value at Risk (VaR)

Problem
Define VaR. What are its limitations?

Solution
Definition. $\text{VaR}(1-\alpha, T)$ is the loss threshold such that $P(\text{Loss} > \text{VaR}) = \alpha$ over horizon T.
Example. "95% 1-day VaR = \$1M" means there is a 5% chance of losing more than \$1M tomorrow.
Parametric calculation. Assuming normally distributed returns with mean μ and volatility σ:

$$\text{VaR}_{95\%} = -\mu + z_{0.95} \cdot \sigma \approx 1.65\,\sigma \quad \text{(for daily horizon, where } \mu \approx 0\text{)}.$$

Limitations:

- **Not coherent**: VaR can violate subadditivity. Combining two portfolios may *increase* VaR.
- **Says nothing about tail**: VaR tells you the threshold, not how bad losses beyond it could be.
- **Assumes Normal**: Real returns have fat tails; parametric VaR underestimates extreme losses.

Alternative. Expected Shortfall (CVaR) addresses these shortcomings:

$$\text{CVaR} = \mathbb{E}[\text{Loss} \mid \text{Loss} > \text{VaR}].$$

It is coherent (satisfies subadditivity) and tail-aware.

Problem 4: Diversification

Problem
Why does adding assets reduce portfolio risk?

Solution
Portfolio variance is:

$$\sigma_p^2 = \sum_i w_i^2 \sigma_i^2 + \sum_{i \neq j} w_i w_j \sigma_i \sigma_j \rho_{ij}.$$

If $\rho_{ij} < 1$, cross-terms partially cancel. In the limit of n equal-weighted uncorrelated assets:

$$\sigma_p^2 = \frac{\bar{\sigma}^2}{n} \to 0.$$

Key Insight: Diversification eliminates **idiosyncratic** risk. Only **systematic** risk (market factor) remains—the "correlation floor" you cannot diversify away.

Problem 5: Sharpe Ratio Under Leverage

Problem
What happens to Sharpe when you lever up?

Solution
Setup. Let μ and σ denote the expected return and volatility of the unlevered portfolio, and let r_f be the risk-free rate. Apply leverage factor $k > 0$ (i.e., invest k dollars per dollar of equity, borrowing $k-1$ dollars at rate r_f).
Levered return. The excess return above the risk-free rate scales linearly with k:

$$\mu_{\text{lev}} = r_f + k(\mu - r_f), \qquad \sigma_{\text{lev}} = k\,\sigma.$$

Sharpe ratio. The leverage factor cancels in the ratio:

$$SR_{\text{lev}} = \frac{\mu_{\text{lev}} - r_f}{\sigma_{\text{lev}}} = \frac{k(\mu - r_f)}{k\,\sigma} = \frac{\mu - r_f}{\sigma} = SR_{\text{unlev}}.$$

Sharpe is invariant to leverage. This is precisely why Sharpe ratio is the canonical risk-adjusted performance measure: any portfolio can be levered or delevered to hit a target volatility, so the Sharpe ratio captures the intrinsic quality of the strategy independent of position sizing.
Caveat: In practice, leverage introduces borrowing costs above r_f and the risk of margin calls, both of which degrade realized Sharpe.

Problem 6: Information Ratio vs Sharpe Ratio

Problem
What's the difference?

Solution
Sharpe Ratio. Let μ_p be the portfolio's expected return, r_f the risk-free rate, and σ_p the portfolio volatility. The Sharpe ratio measures *absolute* risk-adjusted performance:

$$SR = \frac{\mu_p - r_f}{\sigma_p}.$$

A high Sharpe ratio means the portfolio delivers strong total returns per unit of total risk.
Information Ratio. Let α be the portfolio's active return (excess return above a benchmark) and σ_ϵ the tracking error (volatility of the active return). The information ratio measures *relative* risk-adjusted performance:

$$IR = \frac{\alpha}{\sigma_\epsilon}.$$

A high information ratio means the portfolio consistently outperforms its benchmark (e.g., the S&P 500) relative to the volatility of that outperformance.
Relationship. For a market-neutral strategy with no benchmark exposure, the active return equals the total excess return, and tracking error equals total volatility, so $IR \approx SR$.

Problem 7: Kelly Criterion

Problem
What fraction of capital to bet for optimal long-term growth?

Solution
Setup. A bet wins with probability p and loses with probability $q = 1 - p$. The odds are $b : 1$, meaning you win \$$b$ for every \$1 wagered. Let f denote the fraction of your current bankroll bet on each round.
Derivation. The Kelly criterion chooses f to maximize expected log-wealth (equivalently, the geometric mean growth rate). After one bet, your wealth is:

$$W_{n+1} = W_n\big[(1-f) + f(1+b)\big] = W_n(1+fb) \quad \text{if you win,}$$
$$W_{n+1} = W_n(1-f) \quad \text{if you lose.}$$

The expected log-growth per round is:

$$g(f) = p\ln(1+fb) + q\ln(1-f).$$

Setting $g'(f) = 0$:

$$\frac{pb}{1+fb} - \frac{q}{1-f} = 0,$$

which gives the optimal fraction:

$$f^* = \frac{p \cdot b - q}{b} = \frac{p(b+1) - 1}{b}.$$

Example. For a fair coin ($p = 0.5$) with 2:1 odds ($b = 2$):

$$f^* = \frac{0.5 \times 3 - 1}{2} = 0.25.$$

You should bet 25% of your bankroll each round.
Continuous version. For log-normal returns with excess mean $\mu - r$ and volatility σ, the Kelly fraction becomes:

$$f^* = \frac{\mu - r}{\sigma^2}.$$

Key Insight: Kelly maximizes long-term growth but has high short-term volatility. Many practitioners use "half-Kelly" ($f^*/2$) to reduce drawdowns at the cost of a modest reduction in growth rate.

Problem 8: Why Variance, Not Standard Deviation?

Problem
Why do we often work with σ^2 rather than σ?

Solution
Variance has nice mathematical properties:

- **Additivity**: For uncorrelated assets, $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$. Standard deviation doesn't add.
- **Decomposition**: $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$—clean algebra.
- **Portfolio formula**: $\sigma_p^2 = w^T \Sigma w$ is quadratic in weights (nice for optimization).

Intuition: Variance is the natural "squared distance" in return space. It's the language of mean-variance optimization (Markowitz).

7 Special Topics

Problem 1: Turnover Penalty Effect

Problem
What happens as κ increases in $\min_w \frac{1}{2} w^T \Sigma w - \lambda \mu^T w + \kappa \|w - w_0\|_2^2$?

Solution
The objective is:

$$\min_w \ \frac{1}{2}\, w^T \Sigma\, w \ - \ \lambda\, \mu^T w \ + \ \kappa\, \|w - w_0\|_2^2$$

The third term is a quadratic (ridge-style) penalty on the distance between the new portfolio w and the current portfolio w_0. As κ increases:

- The optimal w is pulled closer to w_0, reducing turnover.
- Less of the alpha signal μ is captured, because the optimizer sacrifices return to avoid trading.
- In the limit $\kappa \to \infty$, $w \to w_0$ and no trades are executed.

Intuition: The penalty acts as a ridge shrinkage term centered on the current book. The parameter κ controls the trade-off between alpha capture and transaction cost avoidance.

Problem 2: Woodbury Use

Problem
How do you efficiently update Σ^{-1} after a rank-1 change?

Solution
Suppose the covariance matrix receives a rank-1 update:

$$\Sigma_{\text{new}} = \Sigma + u\, u^T$$

The Sherman–Morrison formula (a special case of the Woodbury identity) gives the updated inverse directly:

$$\Sigma_{\text{new}}^{-1} = \Sigma^{-1} \ - \ \frac{\Sigma^{-1}\, u\, u^T\, \Sigma^{-1}}{1 + u^T \Sigma^{-1}\, u}$$

This requires only matrix-vector products ($O(n^2)$), avoiding a full $O(n^3)$ re-inversion.
Intuition: When you already have Σ^{-1} and the covariance matrix changes by a low-rank perturbation (e.g., adding or removing a factor), the Woodbury identity lets you update the inverse incrementally, which is critical in real-time portfolio optimization.

Problem 3: Linear Cost Threshold (No-Trade Band)

Problem
When is optimal trade size zero?

Solution
Consider an optimizer that trades to capture alpha signal a while paying a linear transaction cost c per unit traded. The net gain from trading one unit is $a - c$.
If $|a| < c$, the cost exceeds the alpha, and the optimal trade size is zero. This creates a **no-trade band**:

$$\Delta w^* = 0 \quad \text{whenever } |a| < c$$

Only when the alpha signal exceeds the cost threshold ($|a| \geq c$) is it profitable to trade.
Intuition: Transaction costs create a dead zone around the current position. Small signals are not worth acting on, because the friction of trading would consume more than the expected gain. This is why real portfolios trade less frequently than a frictionless optimizer would suggest.

Problem 4: Two-Asset Closed Form

Problem
Closed-form optimal weights for two assets?

Solution
For two uncorrelated assets with expected returns μ_1, μ_2 and variances σ_1^2, σ_2^2, the mean-variance optimal portfolio (ignoring the risk-free asset) assigns weights proportional to each asset's return-to-variance ratio:

$$w_i \propto \frac{\mu_i}{\sigma_i^2}$$

Normalizing so the weights sum to one:

$$w_1 = \frac{\mu_1/\sigma_1^2}{\mu_1/\sigma_1^2 + \mu_2/\sigma_2^2}, \qquad w_2 = 1 - w_1$$

Intuition: Each asset's allocation is tilted by its signal-to-variance ratio. An asset with a high expected return but also high variance receives a smaller weight than one might naively expect, because the optimizer penalizes variance.

Problem 5: Impact vs Spread Decomposition

Problem
You buy 1000 shares. Arrival price \$100.00, average fill \$100.15, half-spread 2.5 bps. Break down the slippage.

Solution
Let $p_{\text{arrival}} = \$100.00$ (mid at decision time) and $\bar{p}_{\text{exec}} = \$100.15$ (average fill). The total slippage is:

$$\text{Slippage} = \frac{100.15 - 100.00}{100.00} = 15 \text{ bps}$$

Decompose this into two components:

- **Half-spread cost**: The bid-ask spread is 5 bps, so the half-spread is 2.5 bps. A market buy immediately crosses this spread.
- **Market impact / timing cost**: The remaining $15 - 2.5 = 12.5$ bps reflects adverse mid-price drift during execution (a combination of information leakage, adverse selection, and market impact).

Intuition: Classifying slippage into spread cost and impact cost diagnoses execution quality. Spread cost is largely unavoidable; impact cost is where execution algorithms can add value.

Problem 6: Benchmark Switch

Problem
Your alpha is fast-decaying (intraday). Which execution benchmark: Arrival, VWAP, or POV?

Solution
For a fast-decaying (intraday) alpha signal, the right benchmark is **Arrival Price**.

- **Arrival Price**: Measures cost relative to the price at the moment the trading decision is made. Because the signal decays quickly, execution must be front-loaded to capture it. Arrival price directly penalizes delay, making it the natural match.
- **VWAP**: Spreads execution evenly across the day. This risks missing the alpha window entirely, because by the time later fills occur the signal has already decayed.
- **POV (Percentage of Volume)**: Caps market footprint by tracking participation rate. Useful for large orders in thin markets, but for a fast alpha it is too passive and may miss the move.

Principle: Always align the execution benchmark to the alpha's urgency profile. Fast alpha demands aggressive, front-loaded execution; slow alpha can tolerate patient, spread-minimizing strategies.

Problem 7: Implementation Shortfall

Problem
Arrival price \$100, fill 1000 shares at \$100.15, position closed flat. What is the shortfall?

Solution
Implementation shortfall is the difference between the paper return (at the arrival price) and the actual return (at the execution price). Let $p_{\text{arrival}} = \$100$, $p_{\text{exec}} = \$100.15$, and $q = 1000$ shares.

$$\text{Shortfall} = (p_{\text{exec}} - p_{\text{arrival}}) \times q = (100.15 - 100.00) \times 1000 = \$150$$

Because the position was closed flat, there is no mark-to-close drift component. The entire \$150 is pure execution cost (slippage).
Intuition: Implementation shortfall separates execution slippage from market drift, isolating the cost attributable to the trader's execution quality.

Problem 8: Adverse Selection

Problem
You are quoting (providing liquidity). Order-book imbalance is high and short-term alpha predicts an up move. Should you continue quoting on the offer?

Solution
No — you should **not** continue quoting on the offer (i.e., selling passively) in this scenario.
High order-book imbalance combined with a short-term alpha predicting an up move means:

1. Your resting offer is likely to get lifted (filled) by aggressive buyers.
2. Immediately after you are filled, the price moves up — leaving you short at a worse-than-fair price.
3. The half-spread you earn is overwhelmed by the adverse price movement.

The correct response is to **reduce or cancel your offer** and consider **flipping to the taking side** (buying aggressively) to capture the predicted up move.
Principle: Adverse selection risk overwhelms spread capture whenever informed flow dominates. A market maker must recognize when quoting exposes them to picking off by informed traders.

8 Python & Compute

Problem 1: List vs Tuple

Problem
Key differences?

Solution

- **Mutability**: Lists can be changed (`append`, `pop`); Tuples are immutable.
- **Performance**: Tuples are lighter in memory and faster to create.
- **Usage**: Tuples can be dictionary keys (if they contain only hashables); Lists cannot. Tuples imply structural data (record); Lists imply homogeneous data (array).

Problem 2: Generators

Problem
Explain `yield`. Benefit?

Solution
The `yield` keyword turns a function into a **generator**. When called, the function does not run to completion; instead it returns a generator object that produces values one at a time. Each call to `next()` resumes execution from where it last yielded, preserving all local state.
The primary benefit is **lazy evaluation**:

- A list comprehension builds the entire result in memory — $O(N)$ space.
- A generator yields one item at a time — $O(1)$ space.

This is crucial for processing large log files, streaming data, or representing infinite sequences.

Problem 3: Decorators

Problem
What is `@wrapper`?

Solution
A decorator is a function that takes another function as its argument and returns a new function, typically extending or modifying the original behavior. The `@decorator` syntax is shorthand for `func = decorator(func)`.
Example: A `@timer` decorator wraps a function to measure its execution time. Inside the wrapper, it records `time.time()` before calling the original function, runs it, then prints the elapsed time after it returns.
Best practice: Always apply `@functools.wraps(func)` to the inner wrapper. This copies the original function's name, docstring, and other metadata onto the wrapper, so introspection tools (e.g., `help()`, debugging) still work correctly.

Problem 4: GIL

Problem
What is it?

Solution
Global Interpreter Lock. A mutex that prevents multiple native threads from executing Python bytecodes simultaneously.

- **Impact**: CPU-bound tasks (math) are single-threaded even on multi-core CPUs.
- **Workaround**: Use `multiprocessing` (separate processes, own memory) or libraries that release GIL (NumPy does this for heavy lifting C-routines).
- **IO-bound**: Threading still useful for network/disk waiting.

Problem 5: Pandas Merging

Problem
`inner` vs `left` join?

Solution

- **Inner Join**: Returns intersection of keys. Rows are dropped if key is missing in either.
- **Left Join**: Returns all keys from Left DF. Missing matches in Right DF are filled with `NaN`.
- `pd.merge(df1, df2, on='key', how='inner')`

Problem 6: Vectorized Rolling Z-Score

Problem
$(x - \text{rolling_mean})/\text{rolling_std}$ on 10^6 points (no Python loops).

Solution
Use pandas rolling window methods, which are implemented in C and avoid Python-level loops:

```
rolling_mean = series.rolling(W).mean()
rolling_std  = series.rolling(W).std()

z_score = (series - rolling_mean) / rolling_std
```

The first $W - 1$ entries will be `NaN` (insufficient data for a full window). This approach runs in $O(N)$ time regardless of window size W, because pandas uses an online algorithm internally.

Problem 7: Pandas Groupby + Merge

Problem
Aggregate feature by symbol/day, join back to trades; ensure alignment and missing keys handled.

Solution
Two steps: aggregate, then join the result back to the original trades.

```
# Step 1: compute the per-symbol, per-day aggregate
feat = (df
    .groupby(['sym', 'day'])['x']
    .sum()
    .reset_index(name='x_sum'))

# Step 2: left-join back to the trade-level DataFrame
df = df.merge(feat, on=['sym', 'day'], how='left')
```

Using `how='left'` ensures that trades whose `(sym, day)` key is missing from the aggregation (e.g., due to filtering) receive `NaN` rather than being silently dropped.

Problem 8: Simulate AR(1)

Problem
Estimate ϕ, compute half-life; compare to true; add Ljung–Box to check residuals.

Solution
Step 1 — Simulate the process.

```
eps = np.random.normal(size=N)
x = np.zeros(N)
for t in range(1, N):
    x[t] = phi * x[t-1] + eps[t]
```

Step 2 — Estimate ϕ and compute half-life.

```
phi_hat   = np.corrcoef(x[1:], x[:-1])[0, 1]
half_life = np.log(0.5) / np.log(phi_hat)
```

Compare `phi_hat` to the true ϕ used in simulation; the gap shrinks as N grows.
Step 3 — Check residuals with Ljung–Box.

```
from statsmodels.stats.diagnostic import acorr_ljungbox

resid = x[1:] - phi_hat * x[:-1]
lb_result = acorr_ljungbox(resid, lags=[20])
lb_pvalue = lb_result.iloc[0]['lb_pvalue']
```

If the p-value is large (e.g., > 0.05), the residuals are consistent with white noise and the AR(1) model is adequate.

Problem 9: Rolling Lead/Lag Correlation

Problem
Compute 1-lag cross-corr of two series efficiently (no Python loops) to detect lead-lag alpha.

Solution
Shift one series by one period, then compute rolling Pearson correlation against the other:

```
# a_lagged[t] = a[t-1], so corr(a_lagged, b) measures
# whether yesterday's a predicts today's b
a_lagged = a.shift(1)

rolling_cov = a_lagged.rolling(W).cov(b)
rolling_std_a = a_lagged.rolling(W).std()
rolling_std_b = b.rolling(W).std()

lead_lag_corr = rolling_cov / (rolling_std_a * rolling_std_b)
```

All operations are vectorized (no Python loops). A persistently nonzero `lead_lag_corr` suggests series `a` leads series `b` by one period, which is a potential alpha signal.

Problem 10: Toy Order-Book PnL

Problem
Track slippage vs arrival and VWAP with fees/linear impact; produce a cost breakdown.

Solution
The key cost components are PnL (mark-to-market against arrival), market impact, fees, and slippage. Each is computed vectorized over a fill array:

```
# q            = signed fill quantities (array)
# exec_px      = execution prices (array)
# arrival_px   = arrival (decision) prices (array)
# k            = linear impact coefficient (scalar)
# fee_rate     = per-share fee rate (scalar)

pnl    = np.sum(q * (exec_px - arrival_px))
impact = k * np.abs(q).cumsum()        # cumulative linear impact
fees   = fee_rate * np.abs(q)          # per-fill fee
slip   = (exec_px - arrival_px) * q    # per-fill slippage

report = {'pnl': pnl, 'impact': impact.sum(), 'fees': fees.sum()}
```

Note: The impact model here is simplistic (linear in cumulative volume). In practice, square-root models or transient/permanent decompositions are more realistic.

Problem 11: Intraday Bucket Standardization

Problem
De-mean/de-vol by time-of-day buckets and recombine; confirm mean $\approx$ 0 and unit vol post-standardization.

Solution
Assign each row to a time-of-day bucket, compute per-bucket statistics, then standardize:

```
# Step 1: assign 5-minute buckets based on time-of-day
df['bucket'] = df['ts'].dt.floor('5min')

# Step 2: per-bucket mean and standard deviation
bucket_stats = df.groupby('bucket')['ret'].agg(['mean', 'std'])

# Step 3: join statistics back and standardize
df = df.join(bucket_stats, on='bucket')
df['ret_std'] = (df['ret'] - df['mean']) / df['std']
```

After standardization, each bucket should have mean ≈ 0 and standard deviation ≈ 1. Verify with `df.groupby('bucket')['ret_std'].agg(['mean','std'])`.

9 Brain Teasers

Problem 1: Trailing Zeros in $n!$

Problem
How many trailing zeros in 1000!?

Solution
Each trailing zero corresponds to a factor of $10 = 2 \times 5$ in $n!$. Because factors of 2 are far more abundant than factors of 5, the number of trailing zeros equals the total number of times 5 divides $n!$.
A factor of 5 is contributed by every multiple of 5, an extra factor by every multiple of 25, another by every multiple of 125, and so on. The general formula is:

$$\left\lfloor \frac{n}{5} \right\rfloor + \left\lfloor \frac{n}{25} \right\rfloor + \left\lfloor \frac{n}{125} \right\rfloor + \left\lfloor \frac{n}{625} \right\rfloor + \cdots$$

For $n = 1000$:

$$200 + 40 + 8 + 1 = 249 \text{ trailing zeros}$$

Problem 2: First to 100

Problem
Two players alternate adding 1–10 to a running total starting at 0. First to reach exactly 100 wins. Winning strategy?

Solution
Work backwards from 100. If you leave your opponent at 89, the largest number they can add is 10, reaching at most 99 — so you can always finish to 100 on your next turn.
By the same logic, each "safe" position is 11 less than the previous one. The full chain of winning positions is:

$$89,\ 78,\ 67,\ 56,\ 45,\ 34,\ 23,\ 12,\ 1$$

Strategy: Go first and say "1." After every opponent move k, respond with $11 - k$. This keeps the running total on the winning positions above, guaranteeing you reach 100 first.

Problem 3: Sums of 4s and 5s

Problem
What is the largest n that *cannot* be written as $4a + 5b$ for non-negative integers a, b?

Solution
Check small values by hand. The numbers that *cannot* be written as $4a + 5b$ (with $a, b \geq 0$) are:

$$1,\ 2,\ 3,\ 6,\ 7,\ 11$$

Starting from 12, every integer is representable:

- $12 = 4 \times 3,\ 13 = 4 + 4 + 5,\ 14 = 4 + 5 + 5,\ 15 = 5 \times 3.$

Once four consecutive integers are achievable, every larger integer is also achievable (just add 4). Since 12, 13, 14, 15 all work, every $n \geq 12$ is representable.
The largest unrepresentable value is therefore 11.
General result (Chicken McNugget / Frobenius): For coprime positive integers a and b, the largest integer that cannot be written as $ax + by$ with $x, y \geq 0$ is:

$$ab - a - b = 4 \cdot 5 - 4 - 5 = 11$$

Problem 4: 100 Prisoners, 100 Boxes

Problem
100 prisoners; 100 boxes each containing one unique name (random permutation). Each prisoner may open 50 boxes; all must find their own name or everyone loses. Is there a strategy with nontrivial survival?

Solution
Model the box contents as a permutation π on $\{1, \ldots, 100\}$: box i contains name $\pi(i)$. Every permutation decomposes into disjoint cycles.
Strategy: Prisoner i starts by opening box i. If it contains name $j \neq i$, open box j next, and so on, following the cycle for up to 50 steps. By construction, prisoner i traverses exactly the cycle of π that contains i, and succeeds if and only if that cycle has length ≤ 50.
Survival probability: The team wins if and only if *every* cycle in the random permutation has length ≤ 50. The probability that a uniformly random permutation on 100 elements contains a cycle of length > 50 is:

$$\sum_{k=51}^{100} \frac{1}{k} \approx \ln 2 \approx 0.69$$

Therefore the survival probability is approximately:

$$1 - \ln 2 \approx 31\%$$

This is vastly better than the naive strategy (each prisoner opens 50 random boxes), which succeeds with probability $(1/2)^{100} \approx 0$.
Key takeaway: Exploiting the cycle structure of the permutation converts an essentially impossible task into one with a roughly $1/3$ chance of success.